ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO ROBOTICS

PAPER 1 · PRACTICE C

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Laid out like the DHET first paper. Section A (30): multiple choice, true or false, matching and acronyms · Section B (70): sensors, motors and Arduino basics, Ohm's law and battery calculations, a robot-arm scenario and a line-following robot program to study.

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INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL calculations and write units with the answers.
6. Write neatly and legibly.

SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 The part of a robot that gathers information about its surroundings is the …

A actuator. B sensor. C gripper. D chassis.

1.2 The Arduino Uno board is built around a …

A microcontroller. B hard drive.
C graphics card. D power station.

1.3 Which memory keeps small amounts of data, such as settings, when the power is switched off?

A RAM B LED C EEPROM D cache

1.4 A light-dependent resistor (LDR) changes its resistance according to the amount of …

A sound. B pressure. C heat. D light.

1.5 Which Arduino function sends a PWM signal to a pin?

A analogWrite() B digitalRead()
C delay() D Serial.begin()

1.6 A motor driver, such as the L298N, is needed in a robot because …

A it makes the motor quieter. B the Arduino pins cannot supply enough current for a motor.
C it stores the program. D it measures distance.

1.7 A robot designed to work safely next to people is a …

A gantry robot. B SCARA robot.
C collaborative robot (cobot). D welding robot.

1.8 The binary number 1010 is equal to the decimal number …

A 8. B 12. C 9. D 10.

1.9 Which communication method uses only two wires, called SDA and SCL?

A UART B I2C C SPI D USB

1.10 A pull-up resistor on a push-button input …

A keeps the input HIGH when the button is not pressed. B increases the motor speed.

C changes AC into DC. D cools the board.

(10)

[10]

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Write only 'True' or 'False' next to the question number (2.1–2.5) in the ANSWER BOOK.

2.1 An infrared obstacle sensor detects objects by sending out infrared light and sensing the light that is reflected back.

2.2 A DC motor turns to an exact angle and holds that position without any feedback.

2.3 Components must be soldered onto a breadboard before a circuit will work.

2.4 The setup() function in an Arduino program runs only once, when the board starts.

2.5 A diode connected across a relay coil protects the switching transistor from voltage spikes.

(5)

[5]

QUESTION 3

Choose a description from COLUMN B that matches an item in COLUMN A. Write only the letter (A–J) next to the question number (3.1–3.10) in the ANSWER BOOK.

	COLUMN A		COLUMN B
3.1	Push button	A	Moves in fixed angular steps
3.2	LDR	B	Lets current flow in one direction only
3.3	Buzzer	C	Gives a digital input when it is pressed
3.4	Stepper motor	D	Supplies portable power to a mobile robot
3.5	Microcontroller	E	Resistance changes with the amount of light
3.6	LED	F	Works as an electronic switch or amplifier
3.7	Transistor	G	Makes a sound as a warning
3.8	Diode	H	Connects points on a breadboard
3.9	Jumper wire	I	Reads inputs, runs the program and controls outputs
3.10	Battery pack	J	Gives off light when current flows through it

(10)

[10]

QUESTION 4

Write out the full form of each acronym below.

4.1 IDE

4.2 LDR

4.3 UART

4.4 ADC

4.5 LiPo

(5)

[5]

TOTAL SECTION A: [30]

SECTION B

QUESTION 5

5.1 Explain how an infrared (IR) line sensor helps a robot to follow a black line on a white floor. (4)

5.2 Explain the difference between a servo motor and a stepper motor, and give ONE robotics application of each. (4)

5.3 Explain why the motors of a robot should have their own power supply, but must share a common ground (GND) with the Arduino. (4)

5.4 Describe the steps to upload a program to an Arduino Uno using the Arduino IDE. (4)

5.5 Give FOUR safety rules to follow when soldering the electronic components of a robot. (4)

[20]

QUESTION 6

6.1 Explain what is meant by the torque and the speed of a motor, and why a gearbox is fitted to the motors of a mobile robot. (4)

6.2 An LED that needs 2 V and 20 mA must be connected to a 5 V Arduino pin. Calculate the value of the series resistor needed. (4)

6.3 A 7,4 V robot battery supplies two motors that each draw 0,6 A, and the electronics, which draw 0,3 A.

6.3.1 Calculate the total current drawn from the battery. (2)

6.3.2 Calculate the power supplied by the battery. (2)

6.3.3 The battery is rated at 2 200 mAh. Calculate how long the robot can run before the battery is flat. (4)

[16]

QUESTION 7

A small robotic arm must sort coloured blocks on a table. It has three servo motors (base, shoulder and gripper), a colour sensor mounted above the pick-up point and a START push button. An Arduino Uno controls the arm. The arm picks up each block and places red blocks in one tray and blue blocks in another.

7.1 Why are servo motors suitable for the joints of this arm? (2)

7.2 How does the colour sensor help the arm to decide where to place a block? (2)

7.3 What does the START button do in this system, and how should it be connected to the Arduino? (2)

7.4 Give TWO safety precautions to take when testing the arm. (2)

7.5 Suggest ONE improvement to the arm that uses another sensor, and say what it would do. (2)

[10]

QUESTION 8

A team has built a line-following robot. Two IR sensors under the front of the robot detect a black line. Each sensor gives LOW when it is over the black line and HIGH when it is over the white floor. Two DC motors are driven by an L298N motor driver.

Study the code below and answer the questions that follow.

```
// Line-following robot with two IR sensors and an L298N motor driver
const int leftSensor = 2;    // LOW = black line under the sensor
const int rightSensor = 3;
const int enA = 5;           // left motor speed (PWM)
const int in1 = 6;
const int in2 = 7;
const int enB = 10;          // right motor speed (PWM)
const int in3 = 8;
const int in4 = 9;
const int speed = 150;

void setup() {
  pinMode(leftSensor, INPUT);
  pinMode(rightSensor, INPUT);
  pinMode(enA, OUTPUT); pinMode(in1, OUTPUT); pinMode(in2, OUTPUT);
  pinMode(enB, OUTPUT); pinMode(in3, OUTPUT); pinMode(in4, OUTPUT);
  digitalWrite(in1, HIGH); digitalWrite(in2, LOW);   // left motor forward
  digitalWrite(in3, HIGH); digitalWrite(in4, LOW);   // right motor forward
}

void loop() {
  int left = digitalRead(leftSensor);
  int right = digitalRead(rightSensor);

  if (left == HIGH && right == HIGH) {          // line between the sensors
    analogWrite(enA, speed);
    analogWrite(enB, speed);
  } else if (left == LOW && right == HIGH) {    // line under the left sensor
```

```
29      analogWrite(enA, 0);
30      analogWrite(enB, speed);
31    } else if (left == HIGH && right == LOW) {    // line under the right sensor
32      analogWrite(enA, speed);
33      analogWrite(enB, 0);
34    } else {                                      // both sensors on black
35      analogWrite(enA, 0);
36      analogWrite(enB, 0);
37    }
38    delay(10);
39  }
```

8.1 Why are pins 5 and 10 used for enA and enB? (4)

8.2 Explain what the robot does when only the LEFT sensor is over the black line (lines 28–30), and why this keeps the robot on the line. (4)

8.3 What is the purpose of lines 17 and 18? (4)

8.4 What happens when both sensors detect black (lines 34–36)? Suggest where on a track this could be useful. (4)

8.5 The robot must move more slowly, at a value of 100. State which line must change, and calculate the PWM duty cycle at this value. (4)

8.6 What does digitalRead() return, and why are the two sensor values stored in variables at the start of loop()? (4)

[24]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER